Professional Portfolio



Julia Clarice Uy

BS Geodetic Engineering

University of the Philippines-Diliman

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About Me

I am a Geodetic Engineer specializing in spatial data science, remote sensing, and urban analytics.

My work focuses on leveraging geospatial data to understand Filipino cities; from mapping population density in informal settlements to predicting housing market dynamics using satellite imagery.

I use tools like Google Earth Engine, Python, QGIS, and machine learning frameworks to turn complex spatial datasets into clear, actionable insights. I am driven by the goal of making urban planning, policy, and conservation in the Philippines more evidence-based and equitable.



Education

University of the Philippines Diliman

GWA: 1.6883
Cum Laude

Bachelor of Science in Geodetic Engineering
Graduated: 2026

Thesis

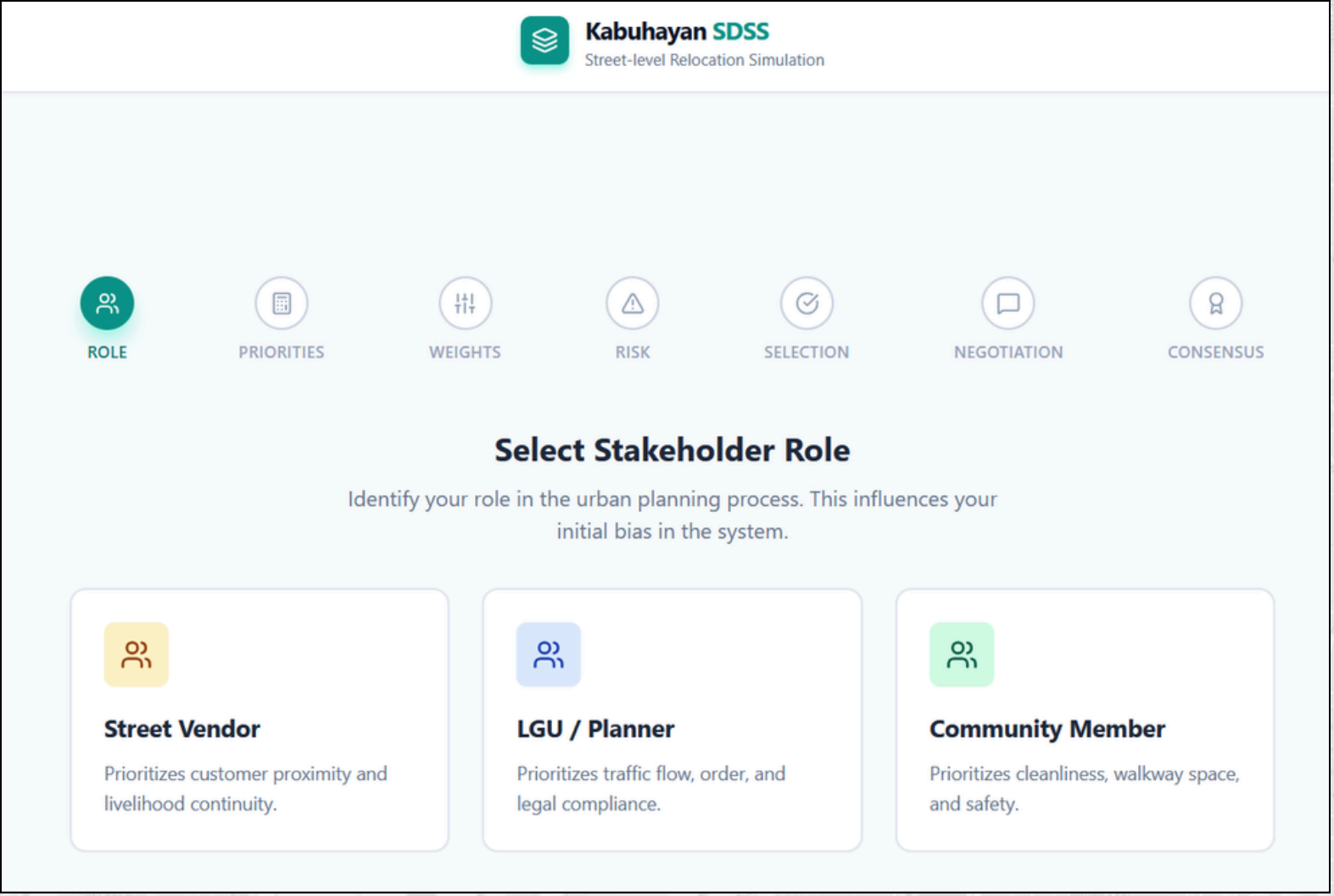
Predicting Urban Housing Rent Using Satellite Imagery and Machine Learning

Relevant Courses

GIS, Remote-sensing, Photogrammetry, Surveying

Class of 2026


GIS/
REMOTE-
SENSING



Vendor Relocation SDSS

OBJECTIVE

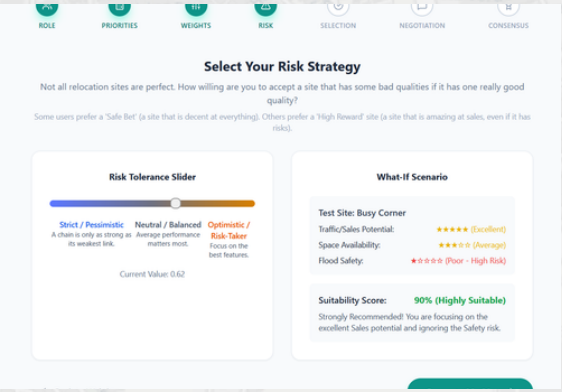
Designed a participatory Web-GIS tool to negotiate and identify mutually acceptable street vendor relocation sites in Manila (https://kabuhayan.netlify.app/).

METHODS:

- Built a React/Leaflet spatial mapping app backed by PostgreSQL/PostGIS to manage stakeholder inputs and layers (flood risk, traffic).
- Implemented an AHP and Borda voting decision workflow to aggregate competing priorities.

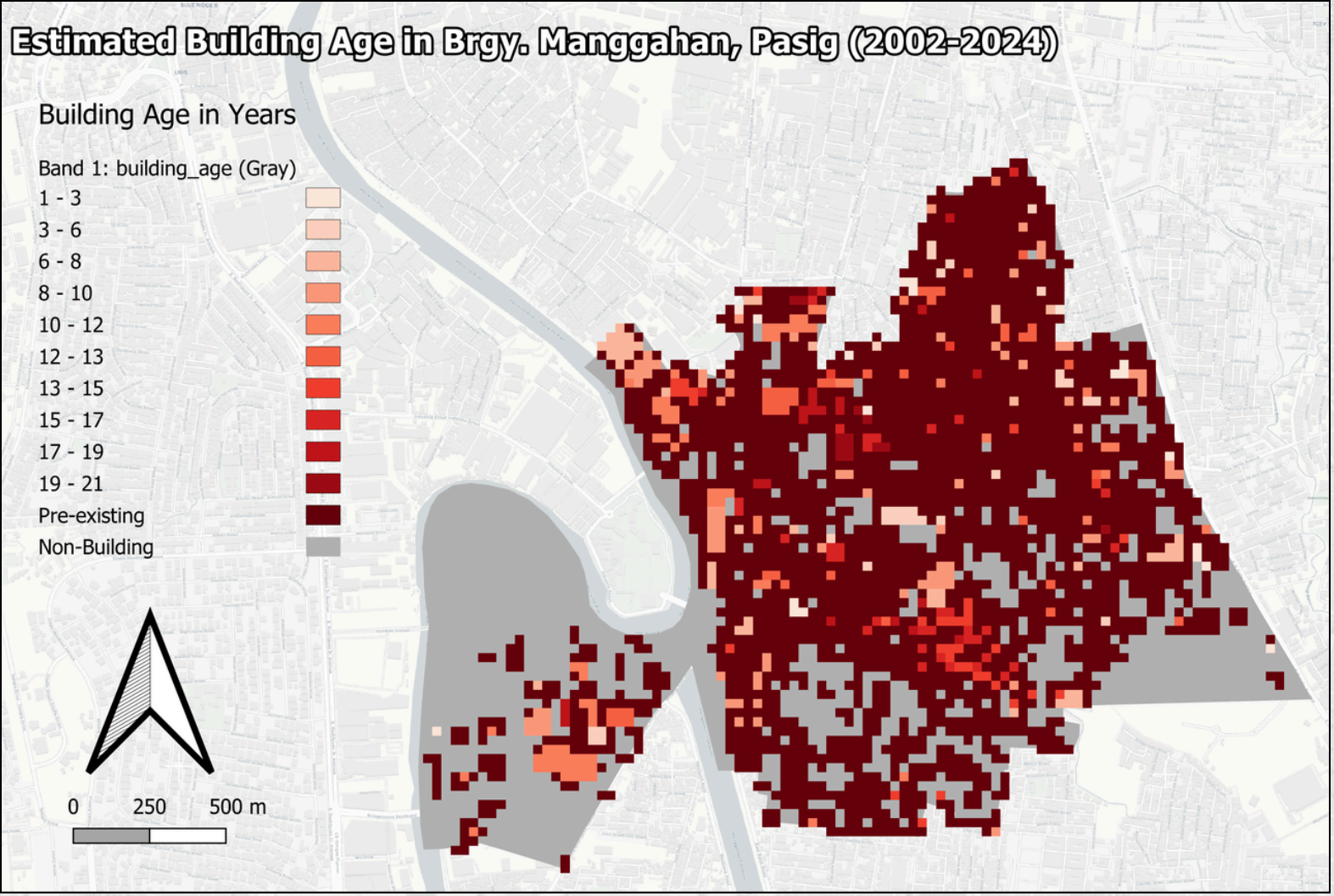
OUTCOME:

Delivered a functional web-map prototype that mathematically balances vendor economic needs with city traffic ordinances to resolve urban planning conflicts.





MY TOOLS:



Urban Housing Rent Prediction, Barangay Manggahan, Pasig City

OBJECTIVE

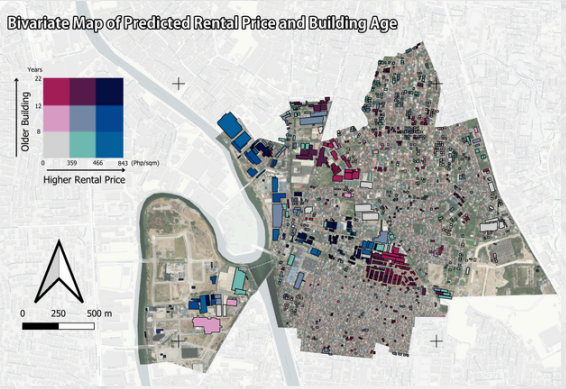
Mapped unrecorded building ages to predict fine-scale rental prices in a mixed-income barangay in Pasig City.

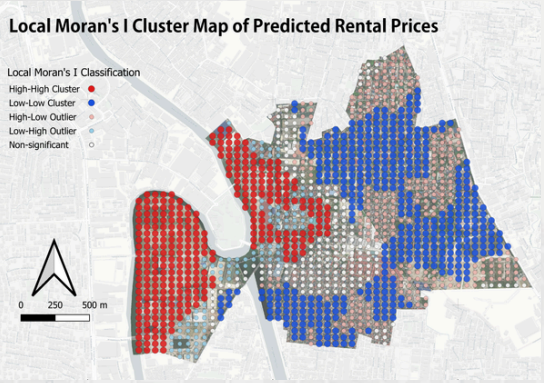
METHODS:

- Built a custom Python geocoder via Google Places API to map web-scraped rental listings.
- Extracted building ages using the LandTrendr algorithm and Landsat time-series in Google Earth Engine.
- Trained an XGBoost machine learning model with SHAP analysis to determine spatial price drivers.

OUTCOME:

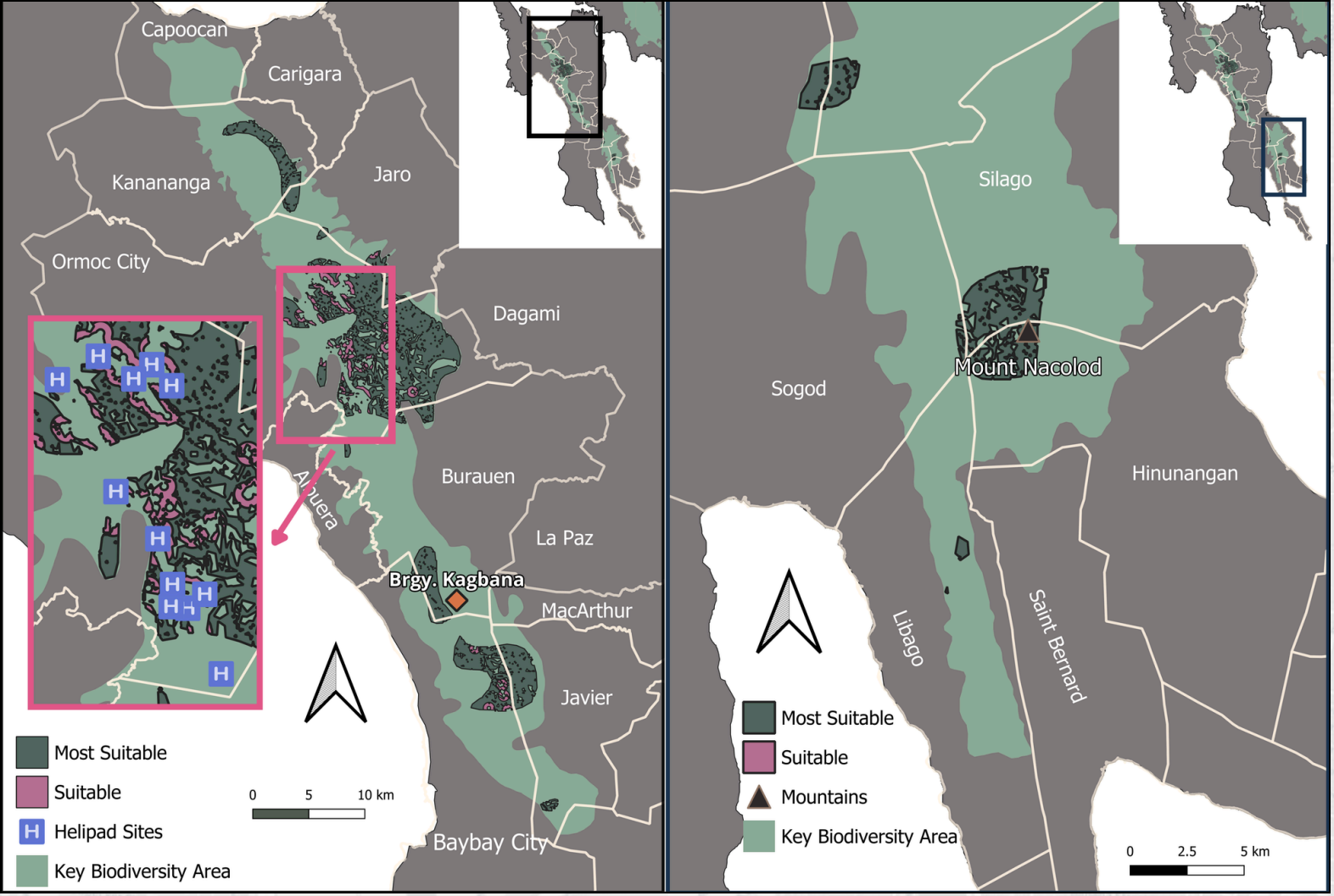
Proved that neighborhood spatial lag drives rent by ₱60/sqm, significantly outperforming individual building age and quality.





MY TOOLS:

ArcGIS



Philippine Eagle Release Sites (Philippine Journal of Science)

OBJECTIVE

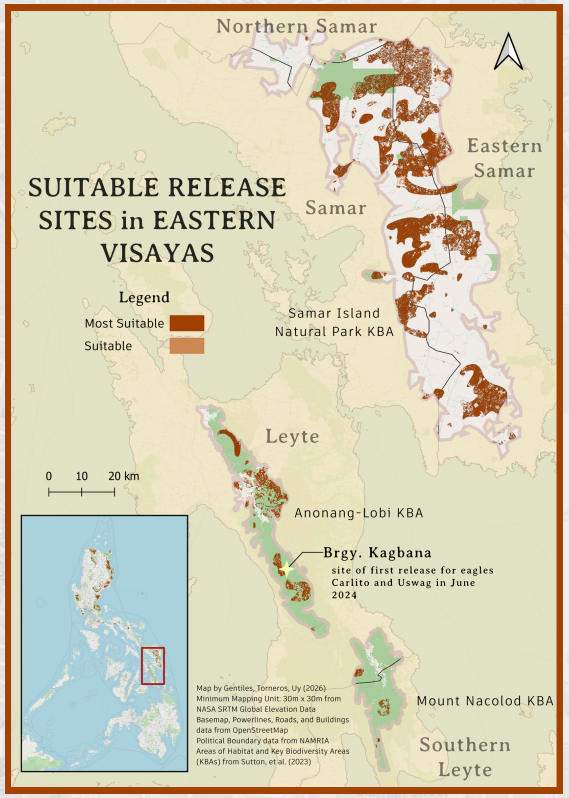
Identified spatially explicit and secure release sites for the reintroduction of critically endangered Philippine Eagles in Leyte.

METHODS:

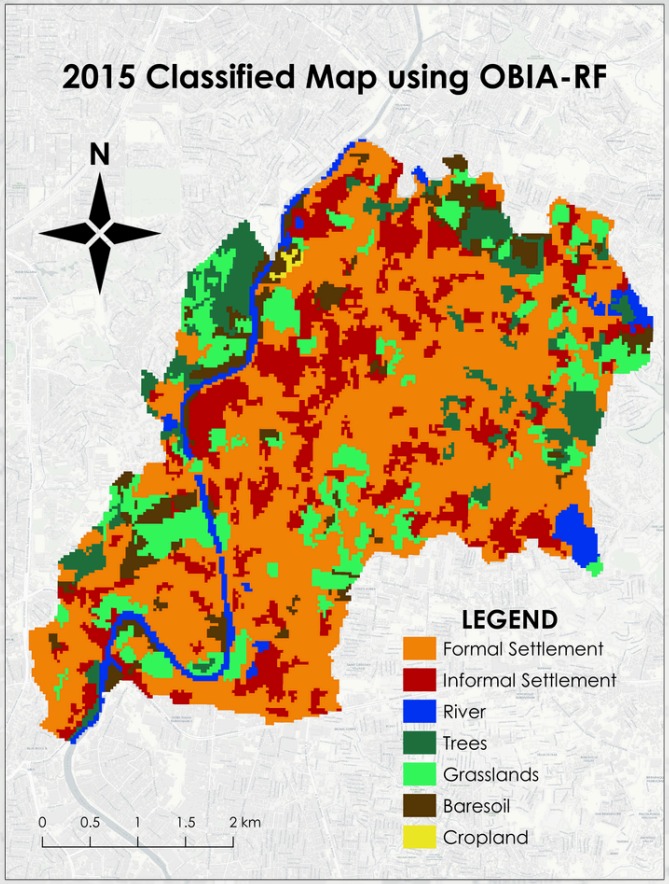
- Converted ecological criteria (slope, powerline buffers, settlement proximity) into spatial constraints.
- Processed environmental datasets (DEMs, Human Footprint Index, Key Biodiversity Areas).
- Applied GIS Multi-Criteria Decision-Making (MCDM) with Boolean overlays to filter out high-hazard zones.

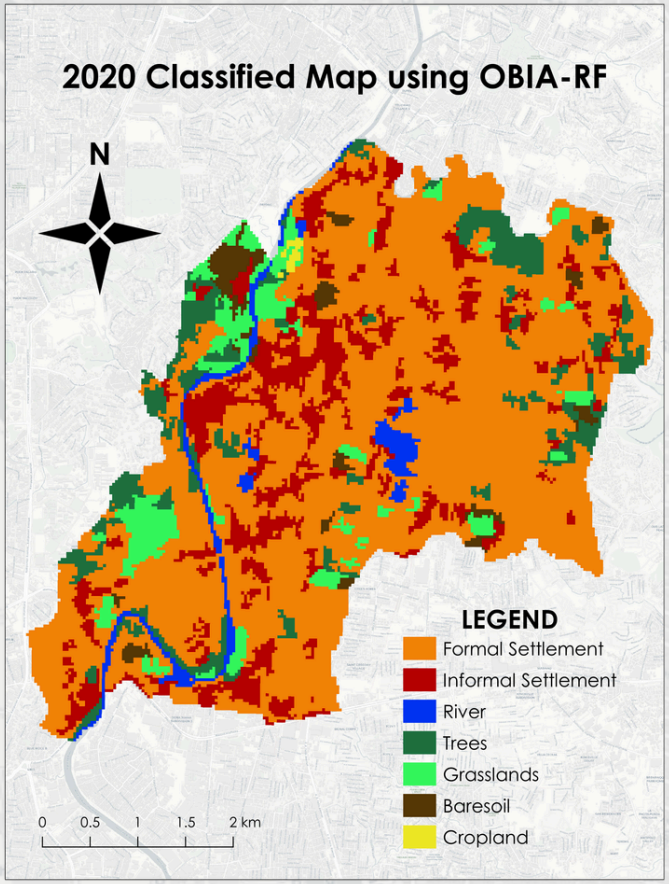
OUTCOME:

Identified 10,320 hectares of suitable habitat, classifying over 9,500 hectares as highly secure release zones located safely away from human infrastructure.



MY TOOLS:





Population Mapping for Informal Settlements (ISPRS Annals)

OBJECTIVE

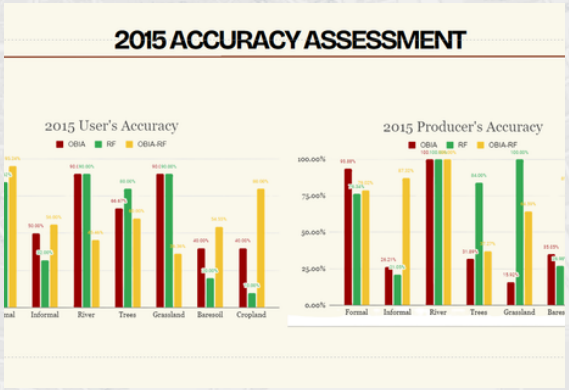
Generated highly accurate, block-level population density maps for formal and informal settlements in Marikina City using satellite imagery.

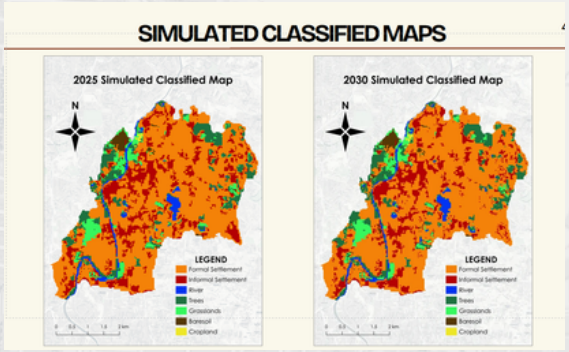
METHODS:

- Classified urban land use from Sentinel-2 imagery by analyzing irregular roof patterns.
- Evaluated Random Forest, OBIA, and Hybrid OBIA-RF models to maximize classification accuracy.
- Downscaled barangay census data into high-resolution 30m grid populations based on settlement patterns.

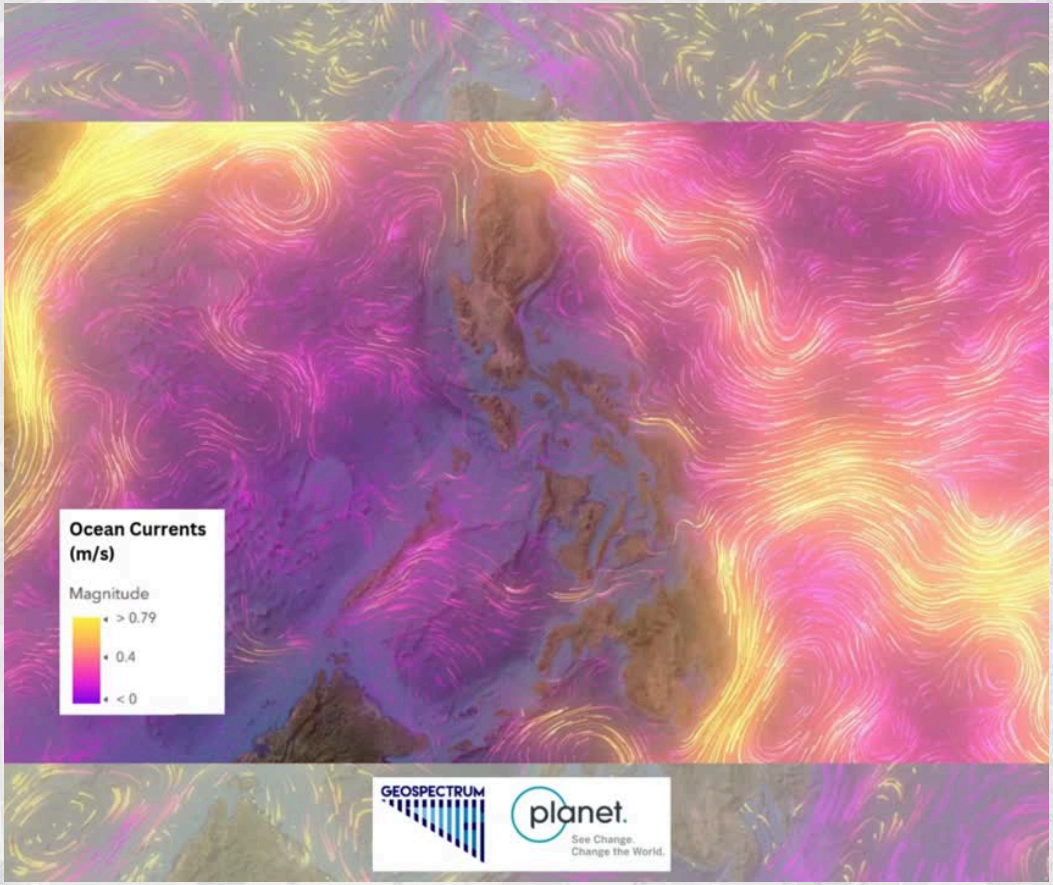
OUTCOME:

Achieved a 72.19% classification accuracy for informal settlements, revealing hidden high-density population clusters that are invisible on standard census maps.





MY TOOLS:



Geospectrum Internship Tasks, 2025

OBJECTIVE

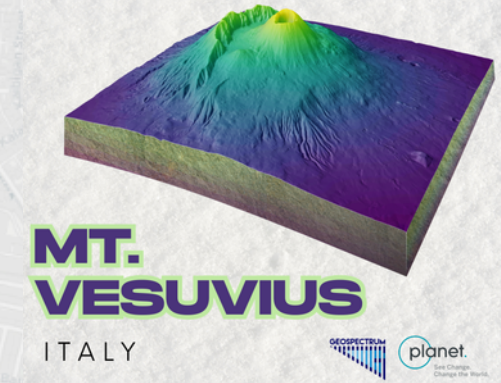
Designed engaging geospatial visualizations for Geospectrum's social media channels. The goal was to highlight company capabilities and communicate complex environmental data to the public.

METHODS:

- Processed HYCOM data to map ocean current magnitudes and flow directions in the Philippines.
- Created high-resolution 3D terrain dioramas (e.g., Mt. Vesuvius) from DEM datasets.
- Used advanced GIS rendering techniques to transform raw scientific data into presentation-ready visuals.

OUTCOME:

Successfully delineated high-turbidity sediment plumes and provided actionable geospatial intelligence for targeted erosion control in rapidly developing coastal zones.

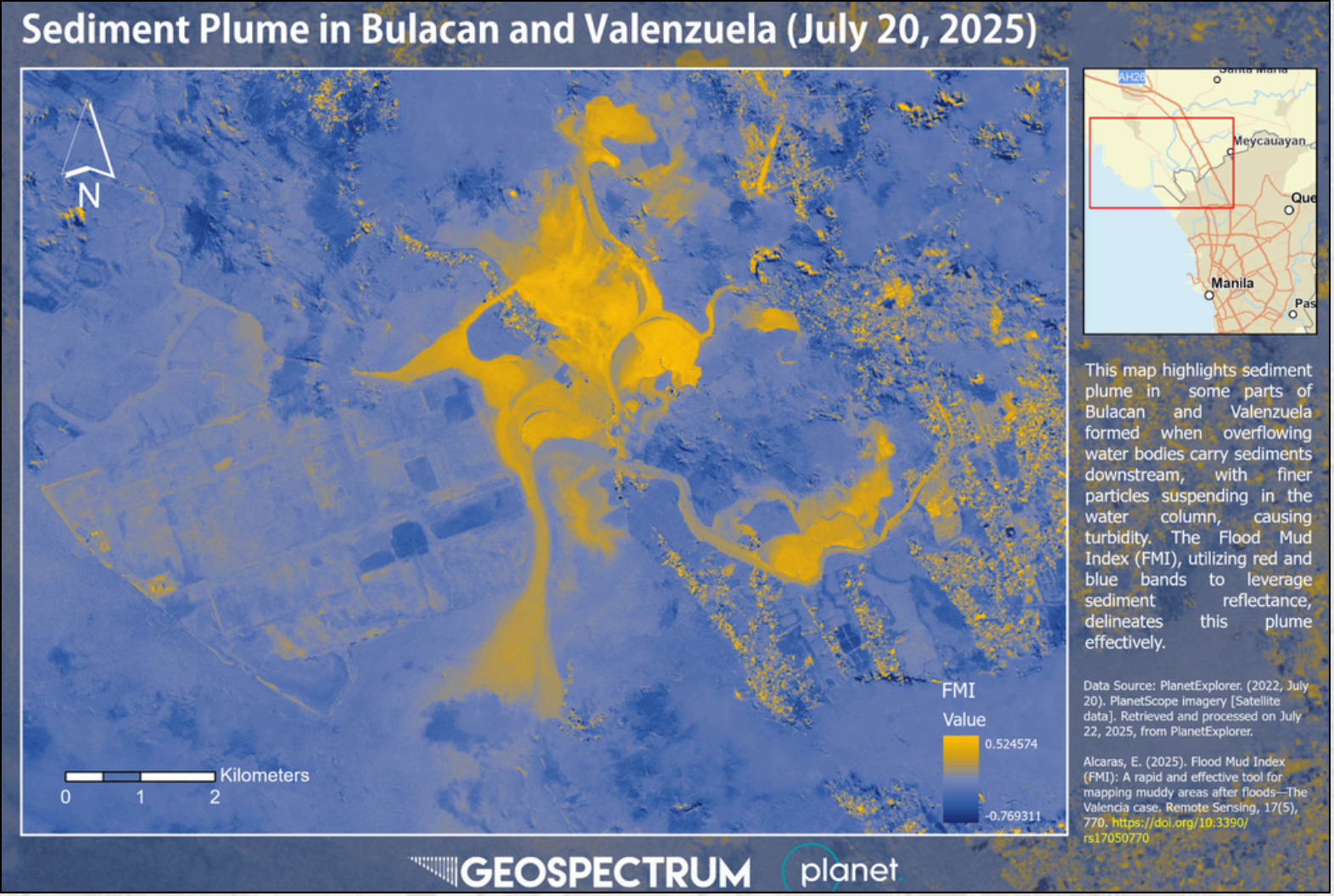






MY TOOLS:

ArcGIS
planet.



Geospectrum Internship Project, 2025

OBJECTIVE

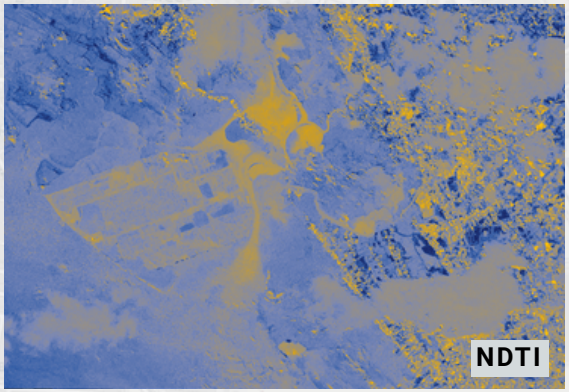
Mapped sediment plumes and muddy floodwaters in Bulacan and Valenzuela to identify critical areas requiring erosion control near the New Manila International Airport.

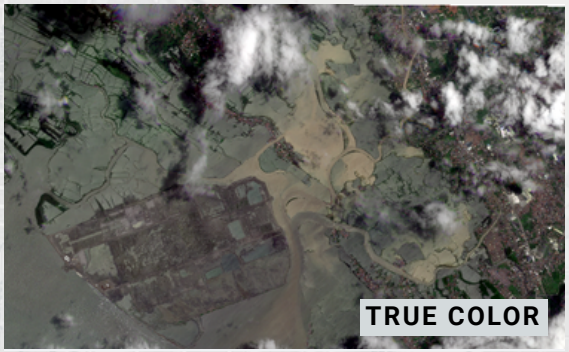
METHODS:

- Processed high-resolution PlanetScope satellite imagery in QGIS during an internship at Geospectrum.
- Leveraged Flood Mud Index (FMI) on red/blue bands instead of NDTI to better separate turbid floodwaters from dry land.

OUTCOME:

Successfully delineated high-turbidity sediment plumes and provided actionable geospatial intelligence for targeted erosion control in rapidly developing coastal zones.





MY TOOLS:

Other Skills:



GIS & Remote-Sensing

QGIS, ArcGIS, ArcMap, OpenStreetMap, SAR



Programming and Machine Learning

Python, Google Earth Engine (GEE), Postgres SQL, R



CAD & Data Entry

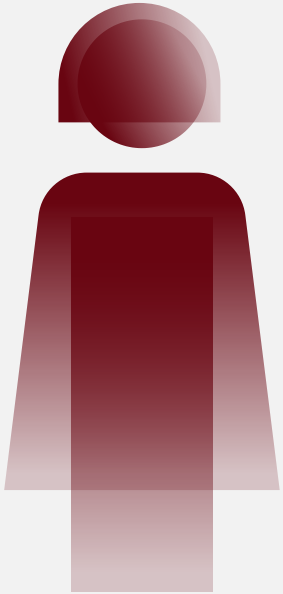
AutoCAD, Civil 3D, Microsoft Excel

Thank You

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